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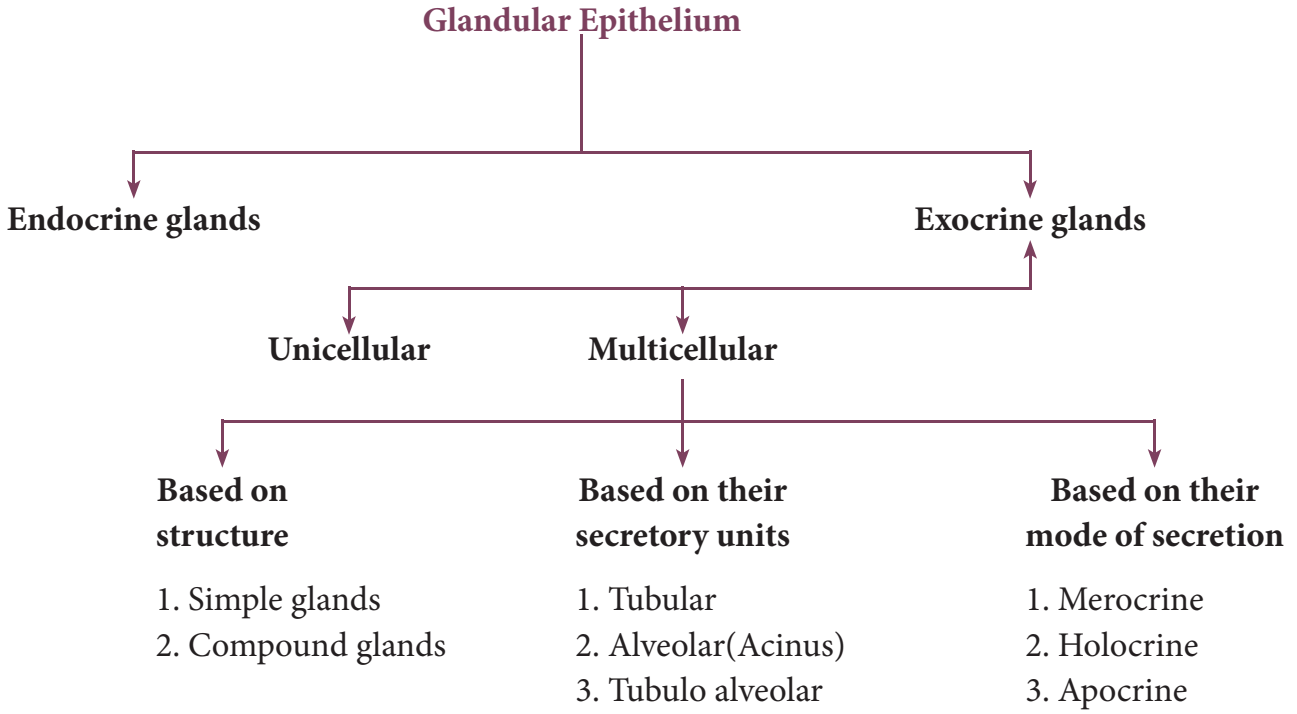
and surface of the ovary. Its main functions are secretion and absorption. The **columnar epithelium** is composed of single layer of tall cells with round to oval nuclei at the base. It lines the digestive tract from the stomach to the rectum. The two modifications of this lining are the presence of **microvilli** on the apical surface of the absorptive cells and **Goblet cell** which secretes the protective lubricating mucus. The functions of this epithelium include absorption, secretion of mucus, enzymes and other substances. If the columnar cells bear cilia on their free surfaces they are called ciliated epithelium. This **ciliated type** propels mucus by ciliary actions and it lines the small bronchioles, fallopian tubes and uterus. **Nonciliated type** lines most of the digestive tract, gall bladder and secretory ducts of glands.

Pseudo-stratified epithelial cells are columnar, but unequal in size. Although the epithelium is single layered yet it appears to be multi-layered because the nuclei lie at different levels in different cells. Hence, it is also called pseudostratified epithelium and its functions are protection, secretion and absorption. Ciliated forms line the trachea and the upper respiratory tract. The non ciliated forms, line the epididymis, large ducts of a glands and tracts of male urethra

Important epithelial tissue disorders:
Eczema, Psoriasis, Epithelial carcinoma and severe asthma

Some of the cuboidal or columnar cells get specialized for secretion and are called **glandular epithelium** (Figure 3.3). They are mainly of two types: unicellular, consisting of isolated glandular cells (goblet cells of the **alimentary canal**), and multicellular, consisting of cluster of cells (**salivary gland**). On the basis of the mode of pouring of their secretions, glands are divided into two categories namely exocrine and endocrine glands. **Exocrine glands** secrete mucus, saliva, earwax, oil, milk, digestive enzymes and other cell products. These products are released